DE-FAKE

Progress Report

In fulfillment of the requirements for the

NU 302 R&D Project

At NIIT University

Submitted by

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Neemrana

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CERTIFICATE

This is to certify that the present research work entitled "De-FAKE" being submitted to NIIT University, Neemrana, Rajasthan, in the fulfillment of the requirements for the course at NIIT University, Neemrana, embodies authentic and faithful record of original research carried out by Anupam Raj of B Tech (Computer Science and Engineering) at NIIT University, Neemrana,. She /He has worked under our supervision and that the matter embodied in this project work has not been submitted, in part or full, as a project report for any course of NIIT University, Neemrana or any other university.

Dr. Vikas Upadhyaya

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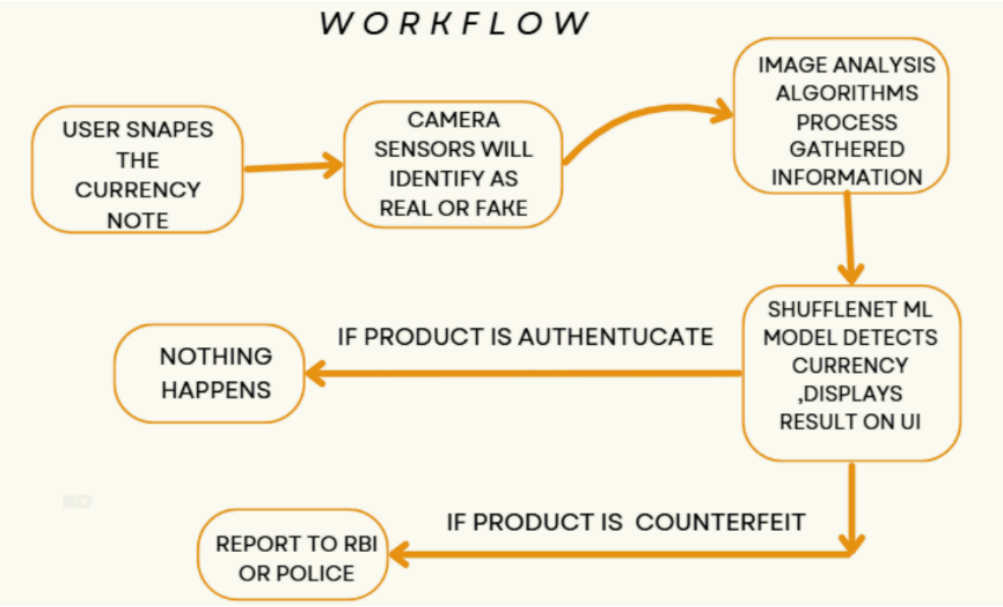

LIST OF TABLES

Table1: Summary of Literature papers:

Paper Title	Methodology	Gaps identified	Key Takeaways
DE-FAKE: Counterfeit Currency Detection	Multi Sensor camera, ORB Feature extraction, Color Histogram, ShuffleNet classification.	Microcontroller not discussed, which one should be used.	Sensor & ML fusion improves accuracy (~98%)
Hyperspectral Imaging for Material Composition Analysis of Banknotes	Multi wavelength spectral imaging for material differentiation.	Expensive & slow while not being portable at the same time.	Spectral features evade Real-time
DeepMoney: Counterfeit money detection using GAN's	GAN Based deep learning for detection	High computation cost .	It improves learning but no hardware integration.
Role of UV Light in detecting Counterfeit notes.	UV fluorescence nature validation	Counterfeit try to create the necessary fluorescence nature by advancement in technology.	Cannot be standalone method.
Counterfeit Detection Pen	Iodine based chemical reaction used to detect starch presence in paper	Advancing technology allows counterfeiters to replicate chemical properties easily of iodine. (Like using of cotton swab)	This can't be a standalone as well has ability to give both false/true positive of counterfeiting.

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INTRODUCTION

Counterfeit currency remains a persistent threat to financial stability, economic security and public security. With advancements in printing technologies, counterfeiters have evolved and public trust. With advancements in printing technologies, counterfeiters have evolved beyond basic visual imitation and now replicate high-resolution patterns, embedded strips, watermark verification, barcode systems, and 2D image-based machine learning models are increasingly insufficient because they rely primarily on visible spectrum features and lack structural depth.

Recent research shows the need for multi dimensional systems that integrate spectral, structural and computational intelligence. The DE-FAKE architecture represents a two layer approach (i) a hardware based sensing unit (ii) a lightweight machine reading unit based on either image or text. The main goal of this research is to strengthen the hardware level by developing a multi sensor ‘Smart hardware camera’ to collect both spectral signatures and 3D geometric features from banknotes in real time.

The aim is to move beyond traditional 2D inspection toward sensor fusion-based counterfeit detection.

LITERATURE REVIEW

Existing literature on counterfeit currency detection can be broadly categorized into four major methodological approaches, each addressing different methods of verifying with the limitation posed or faced while experiments.The development of counterfeit currency detection method has made considerable progress because of new advances in imaging technologies and spectral sensing and machine learning method. The established detection procedure depends on human inspectors who use verification methods to check watermarks and touch based texture evaluation and security thread iodine counterfeit detection pens. The study titled “Analyzing Counterfeit Detection Pen: Limitations and application ” reveals that chemical detection techniques will generate false positive results because counterfeit paper that has been treated can copy the appearance of authentic cotton based materials. The study “The role of UV Light in Detection counterfeit Notes” shows that UV detection systems enhance verification accuracy by showing fluorescent security elements but modern counterfeiters have developed new methods to duplicate UV reaction features which makes these systems less effective. Recent studies have focused on using computer vision and deep learning method .the system described in “DE-FAKE: Counterfeit Currency Detection” uses a mobile platform that combines multi sensor camera technology with ORB feature extraction and color histogram analysis to achieve 93 percent accuracy through classification with its lightweight ShuffleNet model. The system provides real time operational capabilities while maintaining the ability to extend its functions through sensor integration. The system “DeepMoney:Counterfeit money detection using generative Adversarial Networks” uses GAN based learning to enhance feature representation under limited labeled data conditions but these methods need high image quality and stable lighting and enough computational power to work. The study “Hyperspectral Imaging for Material composition Analysis of banknote ” presents hyperspectral imaging techniques. The current system uses 2D image features and single spectral validation methods which do not provide complete 3D geometric measurement that include thickness change and micro embossing assessment. The need for multi sensor hardware systems which build on the DE-FAKE framework to improve structural and spectral validation processes is demonstrated by this deficiency.

A comprehensive summary of key influential papers is in Table 1 [Click Here]

KEY FINDING

Based on our literature reviews and analysis of system, the 5 important discoveries made were:

- Pure 2D vision-based systems are vulnerable to lighting variations and image distortions.
- Spectral sensing (IR + UV) significantly enhances security feature verification.
- Lightweight neural networks such as ShuffleNet enable real-time classification.
- Very limited research incorporates depth profiling or 3D geometric analysis.
- Sensor fusion improves reliability compared to single-modality detection systems.

These findings indicate that combining spectral sensing with structural depth profiling can substantially improve detection accuracy and confidence.

GAPS AND CHALLENGES

Although we discovered multiple advancement through literature review we were also able to point out numerous gaps that still exist in the proposed architecture that have been discussed:

- Absence of method to merge spectral with geometric validation.
- Limited implementation of 3D surface profiling on mobile systems.
- Insufficient real-time multi-sensor synchronization.
- Over dependence upon image data sets that are not calibrated at the hardware level .
- Limited scalability for field deployment.

There were some technical issues that were identified or that came to notice:

- Accurate calibration between LiDAR and ToF sensors
- Real-time data acquisition through MCU
- Handling unseen counterfeit variations
- Signal conditioning for photodiodes
- Power optimization for portable architecture

PROPOSED METHODOLOGY

The proposed system enhances the two-layer DE-FAKE model by strengthening the hardware sensing layer through multi-sensor integration and real-time processing.

5.1 SYSTEM ARCHITECTURE

The system consists of :

Layer 1 - Smart Hardware Camera	Layer 2 - Server-Side Machine Learning Model

The hardware integrates four key sensing mechanisms:

1. Infrared Sensing

   IR LED and photodiode detect hidden mirrored Gandhi texture and IR sensitive print features in the currency notes.

2. UltraViolet Sensing

   UV LED and photodiode verify fluorescent ink responses and security thread authenticity.

3. LiDAR-Based 3D Profiling

   Generates high resolution 3D surface maps to analyze:

● Micro-embossing	● Surface Roughness
● Depth Irregularities	● Geometric Devivation

4. ToF Sensor

   The sensor plays a role in measuring the Note Thickness, distance calibration, Assists LiDA depth accuracy and provides cross validation for structural consistency.

All sensors are controlled through the STM32H743ZI due to its high processing speed, advance timer and DMA support and Real-time data handling capability.

5.2 DATA FLOW

1. Bank-Verified genuine notes are processed for scanning.
2. Sensor data gives a baseline for spectral and structural values.
3. The acquired values become the training dataset for calibration.
4. Fake notes/ or notes which are in doubt are scanned.
5. Data is transmitted to the server.
6. ML Model performs classification.
7. The final confidence score along with the result is displayed on screen.

5.3 DATASET USED

The initial dataset includes the following:

1. Bank verified genuine currency notes.[1]
2. Counterfeit notes. (controlled purpose)

TIMELINES

Task	January	February	March	April
1. Understanding Project	January 15,2026			
2. Literature Survey		February 15,2026		
3. Literature Review		February 28,2026		
4. Implementation I			March 20,2026	
5. Methodology Formulation			March 31,2026	
6. Implementation II				April 15,2026
7. Results & Discussion				April 25,2026
8. Future Roadmap				April 30,2026

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